Jinhee Lee

Website: https://eeleehnij.github.io/ **Email**: jinny6876@gmail.com

RESEARCH INTERESTS

Computer Vision, Vision-Language Models, Robotics, **Generative AI,** and **Bioinformatics**
Focus Areas: Surgical AI, Medical Imaging, Clinical Visual Question Answering, and Human-Robot Interaction

EDUCATION

Ohio State University | Columbus, OH, USA
M.S. in Computer Science and Engineering 2019 - 2021
Courses: Machine Learning, Neural Network, NLP, Advanced Artificial Intelligence, Graphics, Algorithms, and OS

University of Science and Technology | Korea Aerospace Research Institute | Daejeon, Korea
M.Eng. in Aerospace System Engineering 2016 - 2018
Thesis: Study on the Spray Characteristics of Fuel Injectors For Scramjet Engines
Courses: Sensor Signal Processing

Inha University | Incheon, Korea
B.S. in Aerospace Engineering 2010 - 2016
Project: Rocket-Launched Unmanned Aerial Vehicle (UAV) Design and Flight Test of a Scaled Prototype
Courses: Engineering Mechanics, Control Systems, Electromagnetics, Chemistry, Linear Algebra, and Vector Calculus
*Honors: Junior Year Top Student (Full Scholarship) **Military: 2 years of service, Republic of Korea Air Force

PROFESSIONAL EXPERIENCE

CV & AI Engineer | iQ Surgical Inc. | CA, USA 2026 - Present
Surgical Scene Reconstruction and Digital Twin Alignment
- Developing a 3D reconstruction pipeline that mines pixel-level correspondences from surgical video for feature-matching
- Developing a reconstruction-to-Digital-Twin (MRI) registration module, aligning surgical 3D reconstructions with preoperative imaging

AI Research Collaborator | Stanford University | CA, USA 2024 - Present
Surgical AI
- Developed an intraoperative brain surgery segmentation model, improving arachnoid membrane IoU by 7.35%
- Performed intraoperative absolute depth estimation for MVD surgery, achieving <2 mm error
- Developing surgical vision-language foundation models and AI preference alignment

AI Research Collaborator | Pusan National University | Busan, Korea 2026 - Present
Neurosurgical Vision-Language Model
- Exploring vision-language models on intraoperative ICG videoangiography of cerebral aneurysm clipping

AI Research Collaborator | University of Ulsan | Ulsan, Korea 2026 - Present
Ophthalmic OCT Image-to-Image Translation
- Developing an OCT image-to-image generation model predicting postoperative scans from preoperative images

AI Research Collaborator | Bongseng Memorial Hospital | Busan, Korea 2024 - Present

Medical Data Curation and Medical Imaging

- Conducted AI research on brain CT/MRA/MRI for neurovascular classification and brain-lesion segmentation

AI Researcher | Lululab Inc. | Seoul, Korea 2022 - 2026

Segmentation, Detection, and Classification Model Development

- Enhanced facial acne segmentation with GAN-generated synthetic patches, reaching a dice score of 71.47%
- Developed YOLO-based OCR for blood pressure registration for Digital Therapeutics, achieving 94.7% accuracy
- Implemented AI-driven personal color analysis model with 85.8% accuracy and created a demo app

Generative AI

- Developed facial aging and skin disease progression modules using Stable Diffusion model and GAN-based model
- Implemented a face anonymization system leveraging a diffusion model for privacy-preserving medical data
- Developed a ChatGPT-based skin chatbot with GPT fine-tuning, function calling, and RAG

Recommendation Systems

- Designed recommendation systems for cosmetics, skin treatments, and life routines
- Developed a deep learning model for cosmetic ingredients analysis

MS Research Project | Ohio State University | OH, USA 2019 - 2020

Drug Discovery for Alzheimer's Disease

- Collected chemical compound data from PubChem and converted them to SMILES representation
- Performed data vectorization and implemented regression models to identify active compounds for beta-amyloid and tau proteins

Aircraft Engine Researcher | Korea Aerospace Research Institute | Daejeon, Korea 2016 - 2018

- Supersonic combustion and turbojet engine research

Research Intern | Korea Aerospace Research Institute | Daejeon, Korea 2014 Summer

- Aircraft engine compressor research

Aircraft Mechanic | Republic of Korea Air Force | Busan, Korea 2011 - 2013

- Repair and maintenance of the E-737 AEW&C aircraft

PUBLICATIONS AND PRESENTATIONS

[**1**] **Jinhee Lee**, Chan-Young Choi, Huisu Yoon. ***Personal Color Analysis with Vision-Language Models***. [*In revision for resubmission to IEEE Access.]*

[**2**] **Jinhee Lee**, Hwanhee Lee, Jay J. Park, Jeong Woo Ahn, Jong Yun Kwon, Ciara McMahon, Julia Lewandowski, Seohee Park, Sanghoon Lee, and Vivek P. Buch. ***Arachnoid Membrane Segmentation in Intraoperative Microscopic MVD Surgery Scenes.*** In *MICCAI WorkShop on Collaborative Intelligence and Autonomy in Image-Guided Surgery (COLAS)* (pp. 168-177)*,* Springer Nature Switzerland, 2025.

[**3**] **Jinhee Lee**, Chanhyuk Lee, Huisu Yoon, Semin Kim, and Jongha Lee. ***Realistic Skin Trouble Simulation via Image Generation Models***. In *IEEE International Conference On Image Processing (ICIP)*. IEEE, 2025. [**Oral Presentation**]

[**4**] **Jinhee Lee**, Hwanhee Lee, Jay J. Park, Ethan Htun, Bruce Changlong Xu, Sanghoon Cho, Sanghoon Lee, and Vivek P. Buch. ***Intraoperative Absolute Depth Estimation in MVD Surgery***. In *IEEE 38th International Symposium on Computer-Based Medical Systems (CBMS)* (pp. 341-342). IEEE Computer Society, 2025.

[**5**] Semin Kim, **Jinhee Lee**, Chanhyuk Lee, and Jongha Lee. ***Improving Facial Acne Segmentation through Semi-Supervised Learning with Synthetic Images***. *Journal of Korea Multimedia Society,* 27(2), 241-249, 10.9717/kmms.2024.27.2.241, 2024.

[**6**] **Jinhee Lee**, Huisu Yoon, Semin Kim, Chanhyuk Lee, Jongha Lee, and Sangwook Yoo. ***Deep learning-based skin care product recommendation: A focus on cosmetic ingredient analysis and facial skin conditions***. *Journal of Cosmetic Dermatology*, *23*(6), 2066-2077, 2024.

[**7**] **Jinhee Lee**, Sanghoon Lee, Kyungjae Lee, Jaiho Kim, and Soo-Seok Yang. ***Study on Spray Visualization and Atomization Characteristics of Air-assist Type Injector for Scramjet Engine***. *Journal of the Korean Society of Propulsion Engineers*, *21*(5), 88-96, 2017.

TEACHING AND MENTORING EXPERIENCE

Mentoring | University of Ulsan 2026

- Mentored 3 undergraduate students on a dermatology vision-language model project

Mentoring | Lululab Inc. - Sungkyunkwan University 2022 - 2024

- Mentored 9 undergraduate students across 3 AI internship projects: OCR, chatbot, and personal color analysis

PATENT

[**1**] Method and Apparatus for Recommending Cosmetics Suitable for Users Based on the Efficacy of the User's Skin Type and Cosmetic Ingredients, Korean: *10-2022-0079131/10-2450424,* PCT: *PCT/KR2023/008926*

TECHNICAL SKILLS

Programming Languages: Python (Advanced), Java (Intermediate), C++ (Intermediate), MATLAB (Intermediate), C#
Software and Tools: PyTorch, OpenCV, Git, Linux, Docker, Unity Engine, Android Studio, Blender, 3ds Max, LaTeX